Propagation of error

Sections 3.1 and 3.2

1

Parts of Error

We can think of the error of the measurement as being composed of two parts:

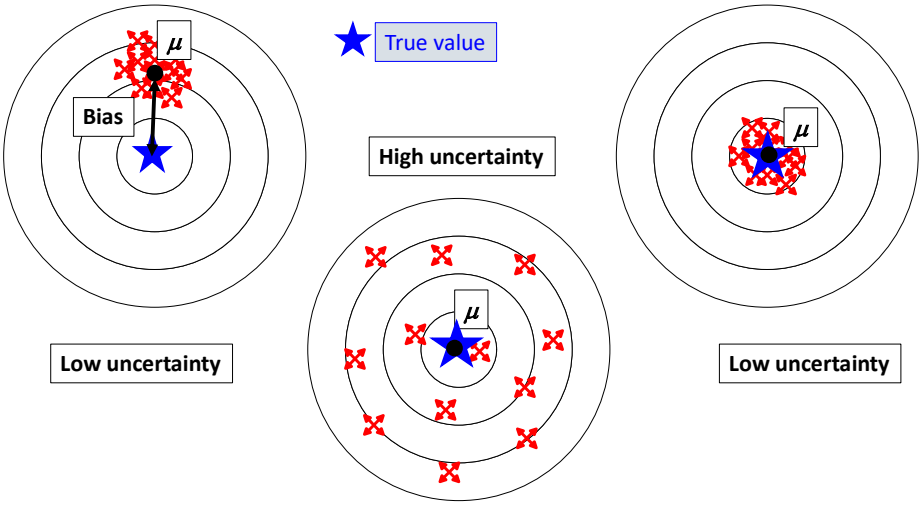
- **Systematic error (bias) - accuracy**
  Bias is the part of the error that is the same for every measurement. For example, a scale that always gives you a reading that is too low.

- **Random error - precision**
  Random error is error that varies from measurement to measurement and averages out to zero in the long run.

Measured value = true value + bias + random error

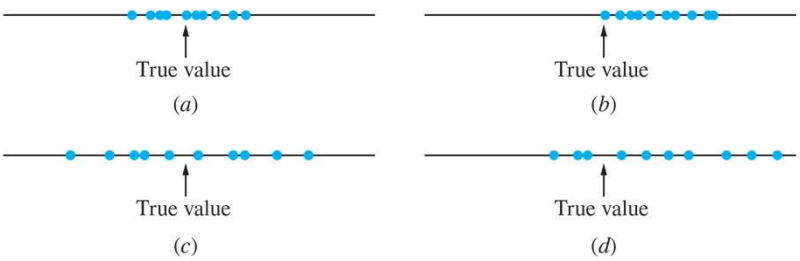
2

Illustration of bias and uncertainty



3

In terms of samples



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Error

A measured value is a random variable with mean μ and standard deviation σ.

The bias in the measuring process is the difference between the mean measurement and the true value:

$$\text{Bias} = \mu - \text{true value}$$

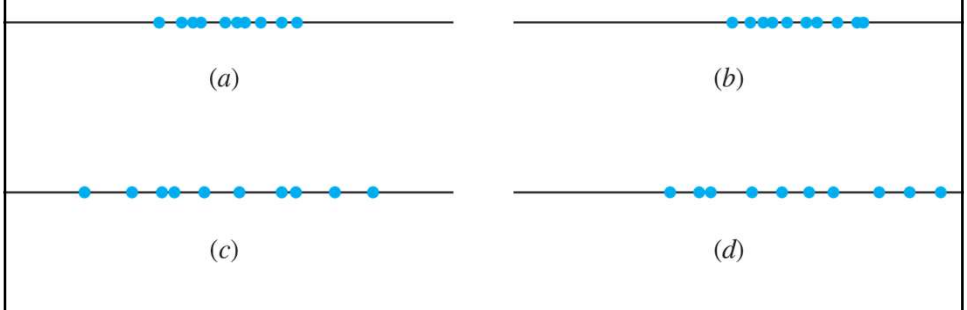
The smaller the bias, the more accurate the measuring process.

The smaller the uncertainty, the more precise the measuring process.

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More on Error

Uncertainty can be estimated from repeated measurements, but in order to estimate the bias, we must have additional information about the true value.



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More on Error

Let $X_1, \cdots, X_n$ be independent measurements, all made by the same process on the same quantity. The sample standard deviation σ can be used to estimate the uncertainty.

Estimates of uncertainty are often crude, especially when based on small samples.

If the true value is known, the sample mean can be used to estimate the bias:

$$\text{Bias} \approx \bar{X} - \text{true value}$$

If the true value is unknown, the bias cannot be estimated from repeated measurements.

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Example

A laboratory sample of gas is known to have a carbon monoxide (C O) concentration of 50 parts per million (ppm).

A spectrophotometer is used to take five independent measurements of this concentration. The five measurements, in ppm, are 51, 47, 53, 53, and 48.

Estimate the bias and the uncertainty in a spectrophotometer measurement.

8

Describing the measurements

When there is no bias:

Measured value ± σ

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Example

The spectrophotometer has been recalibrated, so we may assume that the bias is negligible. The spectrophotometer is now used to measure the C O concentration in another gas sample. The measurement is 55.1 ppm with a standard deviation of 2.8. How should this measurement be expressed?

55.1 ± 2.8

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Concepts to be repeated in different contexts

We will see the same concepts in point estimation as well.

- **Systematic error (bias) - accuracy**
- **Random error - precision**

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